
LiDAR Crime Scenes for VR

System Analysis and Design Report
COMP3018 - Professional Experience
Western Sydney University

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Executive Summary

Based on a proof-of-concept for the Microsoft HoloLens 2, the executive summary describes the creation of a sophisticated crime scene investigation application for the Meta Quest 3. Using the passthrough and depth sensor features of Meta Quest 3, the project's main goal is to improve user collaboration by allowing numerous users to engage within shared virtual crime scenes. Local file loading and rendering of FBX/OBJ files, spatial alignment based on QR codes, multi-user session management with real-time synchronization through Proton, and crime scene manipulation for usability in limited physical spaces are some of the key features. Furthermore, Quest 3's mixed reality elements are integrated to provide excellent visual fidelity and user safety while exploring. In order to ensure alignment with the project goals, the report offers a thorough framework for the application's design, architecture, and testing. After the project is finished, all digital assets and intellectual property (IP) produced will be given to The Mobile Apps Man and ForensicXR, providing a solid basis for useful forensic applications. This synopsis lays the groundwork for an influential and cutting-edge system by condensing the overarching goal, technological underpinnings, and deliverables.

Introduction

This project "ForensicXR: LiDAR Crime Scene Reconstruction System Based on Meta Quest 3" aims to build an immersive multi-user mixed reality platform that generates high-quality model data by scanning crime scenes with high precision and allows users to load and operate in the XR operating environment. The project is based on the previous generation HoloLens upgrade and expansion on the Meta Quest 3/3s platform. Different from the AR scene featured by HoloLens, it makes up for the lack of visual effects of light intensity and computing modules. Meta Quest makes full use of its Passthrough visual perspective and depth sensing capabilities to achieve seamless integration of real and virtual scenes.

The system uses Unity as the development platform, C# programming language, and Photon network framework to achieve cross-device synchronization and multi-person scenes. Users can load LiDAR scan data in OBJ/FBX format locally, anchor the scene through virtual coordinates, and manipulate and scale the scene in real time with other users in a shared space.

We mainly discussed several system development components, including system risks and problem identification: some in-depth analysis of key risk factors such as XR device compatibility, network synchronization stability, model loading performance, and proposed corresponding mitigation strategies; system constraints: covering actual limitations in technology, platform and hardware, such as Unity Personal performance, file format compatibility, and multi-user session alignment mechanism; software architecture diagram: showing the system's functional modules, network communication and data flow structure, and clarifying the client-server communication and resource management process; use case diagram and extended use case: defining the complete user behavior logic from welcome interface, session management, file loading to scene anchoring and interaction; sequence diagram: detailed description of the event interaction and response process between each functional module to ensure a clear logical execution path; entity relationship diagram: establishing the system's core data structure, such as entities such as users, sessions, scene files and their relationships; interface design diagram: presenting the system's key UI interfaces, including the welcome page, main menu, scene joining list and virtual on-site display.

Risks and Issues Related to the System

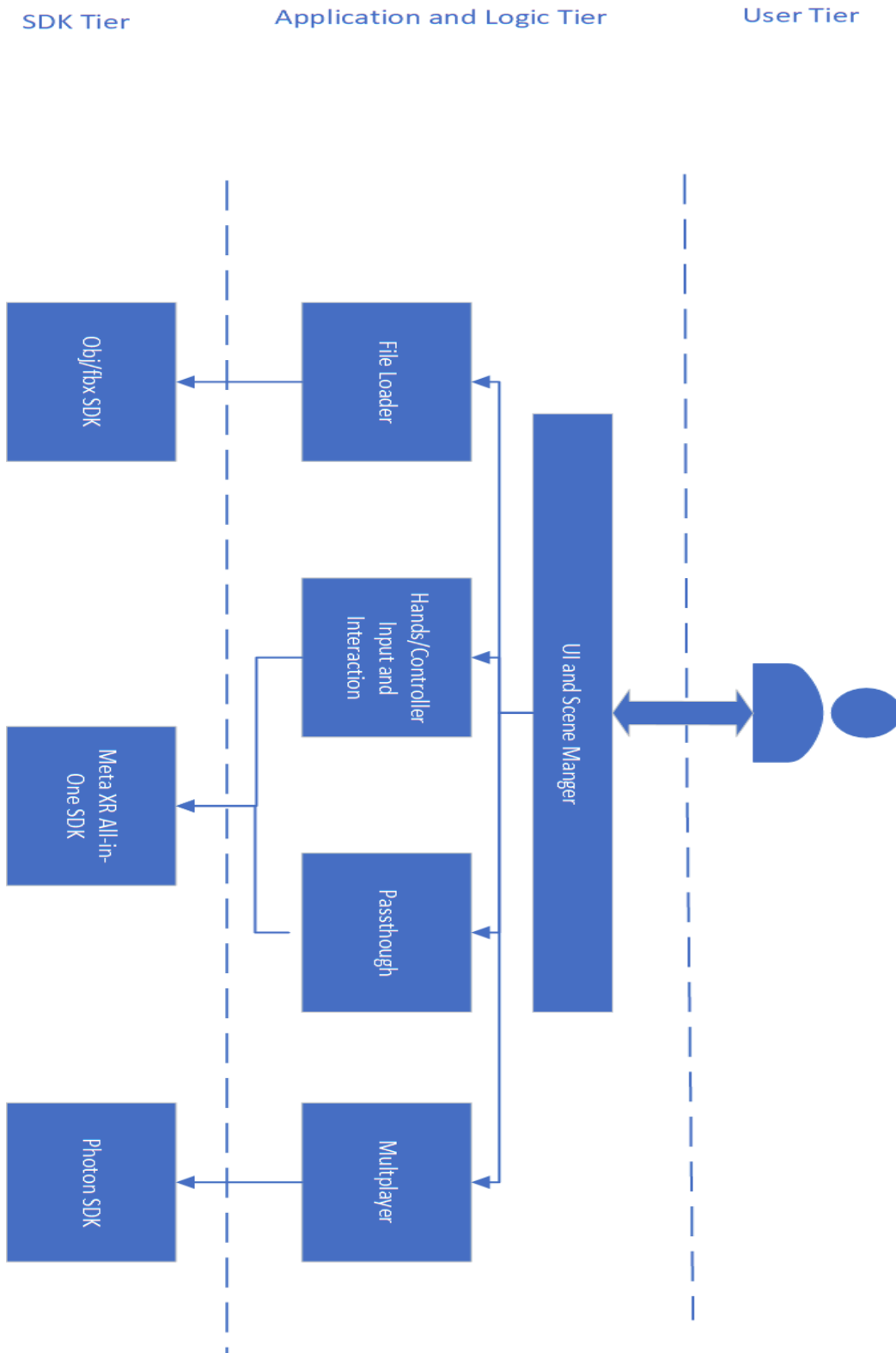
Risk / Issue	Impact Level	Type	Mitigation Strategy
XR software and hardware compatibility issues: For example, Quest 3 does not fully support some Unity development functions	High	Technical	Use the SDK version recommended by Meta in the early stage of development to help develop each functional module and improve compatibility on devices
The team lacks experience in developing integrated Passthrough and LiDAR	Medium	Resource / Knowledge	Refer to Meta/Unity plugin examples, analyze examples, and complete prototypes in stages
Multi-person synchronous communication (such as Photon) is unstable under network delay/packet loss environment	High	Technical	Design a disconnection reconnection mechanism
The FBX/OBJ LiDAR scene model is too large, loading slowly or even crashing	Medium	Performance	Optimize models (reduce faces/LOD), compress textures, use asynchronous loading
XR field of view occlusion or depth error causes 3D model placement deviation	Medium	UX / Technical	Provide model adjustment tools, design scene calibration process, and rescan based on HMD coordinates (such as plane calibration/center point confirmation)
The project cycle is short and the functional complexity is high	High	Project	Phased goals + lock priority features (MVP), adjust scope appropriately
APK size or permission issues affect deployment to Quest 3	Medium	Deployment	Optimize and streamline resources, test and run on the simulator (without a headset)
Risks of field data import (model format)	Low	Data	Add format verification and exception handling mechanisms to limit the source of uploaded data

Poor XR user experience (dizziness, overlap, mapping errors)	Medium	UX	Optimize object scaling and re-set coordinates of headsets to match ground logic
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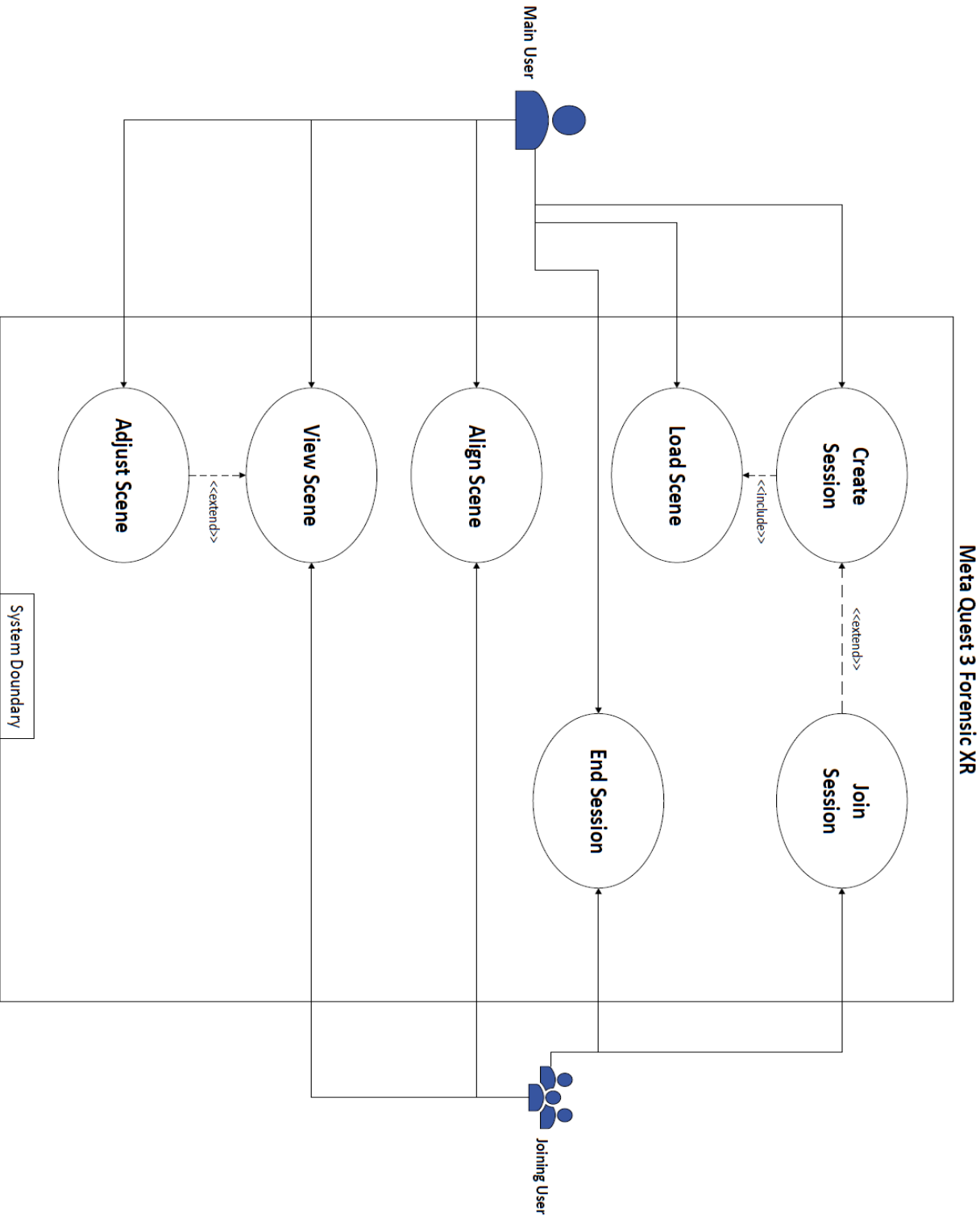
Constraints Related to the System

	Constraint	Mitigation Strategies
1	Safe Navigation with Passthrough	Optimize Quest 3's depth sensing to maintain user safety and prevent collisions in real-world spaces.
2	File Format Compatibility(FBX/OBJ)	Integrate flexible file-parsing mechanisms to ensure FBX/OBJ formats load seamlessly across devices.
3.	Unity Free Version Performance	Analyze the app performance under Unity Personal's constraints, finding ways to optimize the experience
4.	HoloLens to Meta Quest 3 Adaptation	Test the Quest 3 SDK thoroughly while referencing the Hololens 2 application to ensure compatibility.
5.	Multi-User Shared Session Alignment	Using robust qr code scanning to ensure precise scene alignment for all participants.

Software Architecture Diagram



Use Case Diagram



Expanded Use Cases

ID: F001	Feature: Welcome Screen	Priority: Essential
BF1: Session Management		Estimate: 6 hours
Story: <i>As a user, I want the application to display a welcome screen followed by options to create or join a session, so that I can easily begin or enter a collaborative investigation.</i>		
Acceptance Criteria: 1. Welcome Screen: ○ The welcome screen with ForensicXR logo displays on app launch. ○ The screen remains visible for a set amount of seconds before automatically progressing. 2. Session Selection: ○ User sees two clear options: "Create Session" and "Join Session". ○ Options are intuitive and easy to select.		

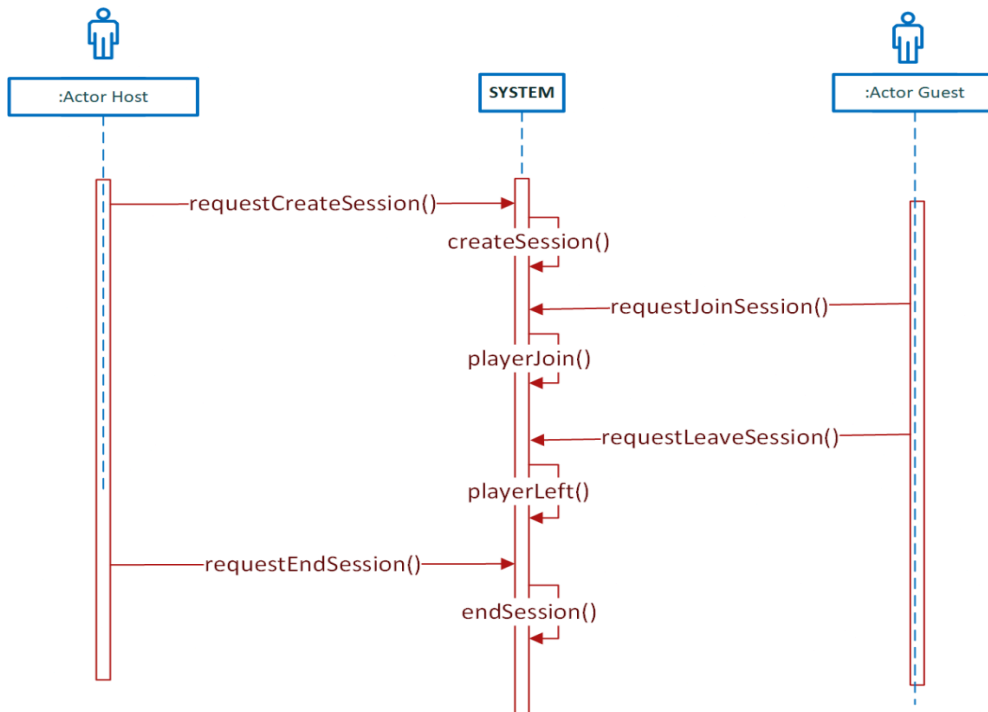
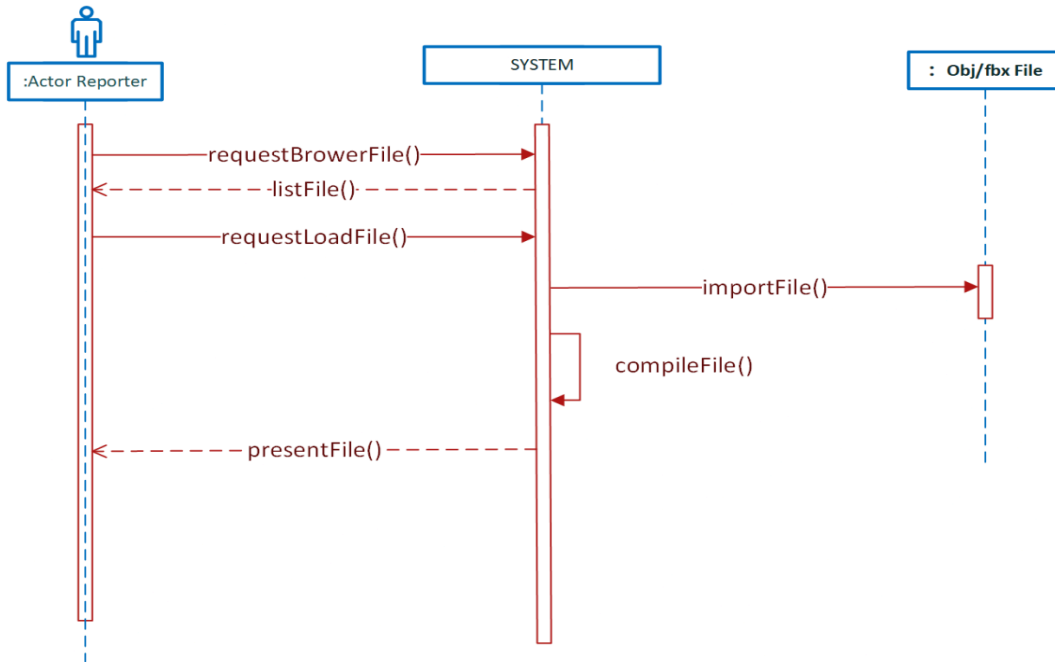
ID: F002	Feature: Create Session & Scene Loading	Priority: Essential
BF2: Scene Management		Estimate: 10 hours
Story: <i>As a user, I want to choose whether to create a new session and load a selected 3D scene file or join an existing session via Proton, so that I can collaborate with other investigators.</i>		
Acceptance Criteria: 1. Create New Session: ○ Selecting "Create Session" generates a unique session ID. ○ Upon session creation, a unique session ID is automatically generated and displayed. ○ The system confirms session creation with a success message. 2. File Selection & Scene Loading ○ Prompt user to select a file (FBX/OBJ) from the device. 3. Join Existing Session: ○ Users can join an existing session by entering a valid room ID.		

ID: F003	Feature:	Priority: Essential
BF3: Scene Interaction		Estimate: 6 hours
Story: <i>As a user, I want to scan a QR code or marker to set the anchor of the scene, ensuring that all participants view the scene in the same location.</i>		
Acceptance Criteria: 1. QR Code Functionality: ○ A “Scan Marker” is prompt in the session interface. ○ The Quest 3 camera detects and successfully scans the QR code/marker. 2. Origin Setting: ○ Upon a successful scan, the system repositions the scene anchor to the marker’s location. ○ All users in the session see the same scene alignment.		

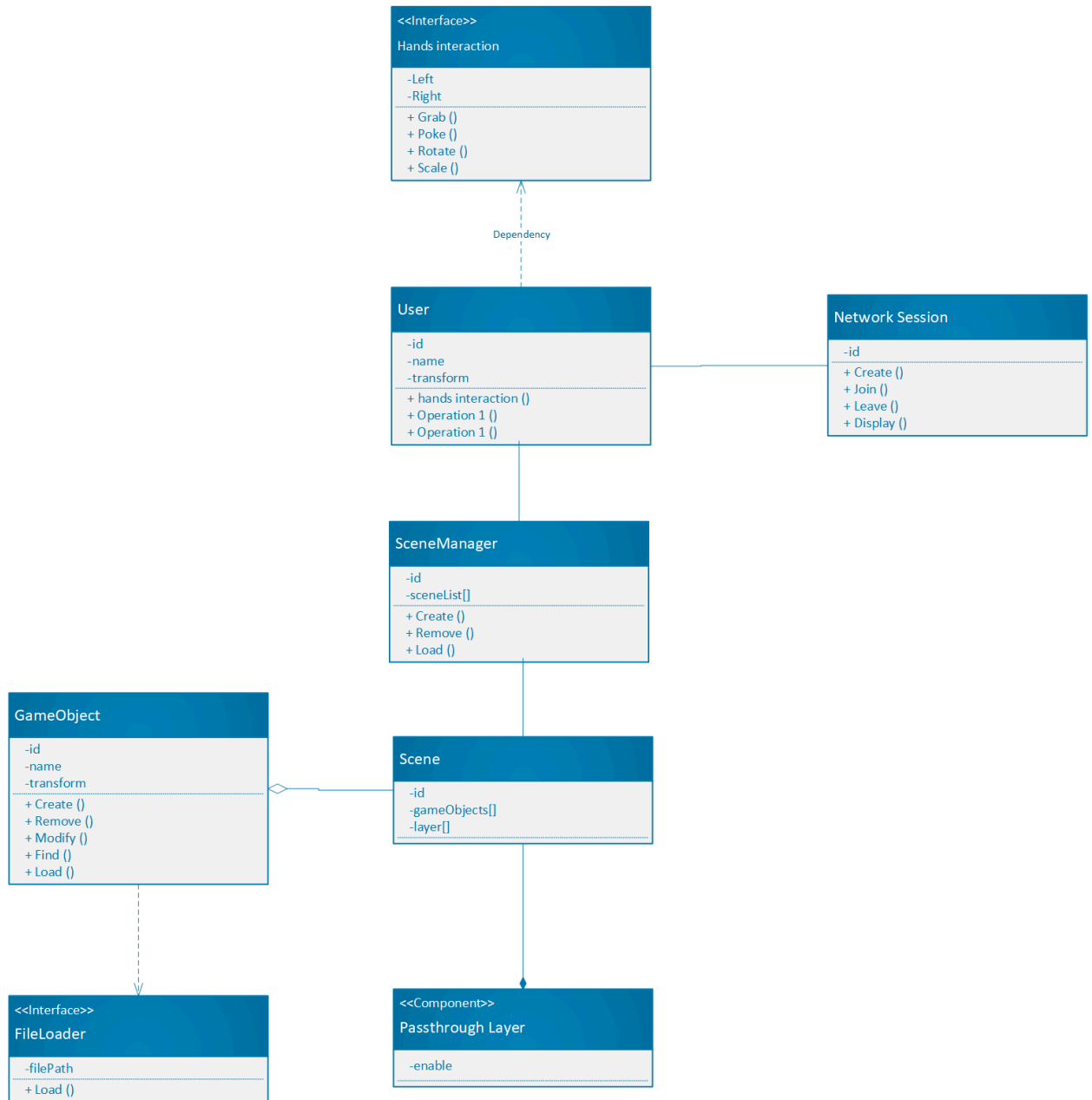
ID: F004	Feature: Scene Loading & Interaction	Priority: Essential
BF4: Scene Management		Estimate: 12 hours
Story: <i>As a user, I want the system to load and spawn the scene from the selected file and allow me to move or rescale it, so that I can adjust the scene to my space during a shared session.</i>		
Acceptance Criteria: 1. Scene Loading: ○ The system successfully loads the selected OBJ/FBX file. ○ The scene loads without errors. 2. Scene Interaction: ○ Users can move and rotate the scene. ○ Users can scale the scene up or down to fit their environment. 3. Real-Time Synchronisation: ○ Any movement, rotation, or scaling performed by the Main User is synchronised in real time with joining users.		

ID: F005	Feature: Passthrough	Priority: Desirable
BF5: User Safety & Environment Awareness		Estimate: 10 hours
Story: <i>As a user, I want to use Meta Quest 3's passthrough and depth sensing features while interacting with the scene, so that I can navigate my physical environment safely.</i>		
Acceptance Criteria: 1. Passthrough Mode: a. The system allows the user to toggle passthrough mode on and off, enabling a view of the real-world surroundings. b. The transition between VR mode and passthrough mode is smooth and reliable.		

Sequence Diagram



Entity Relationship Diagram



Screen Designs

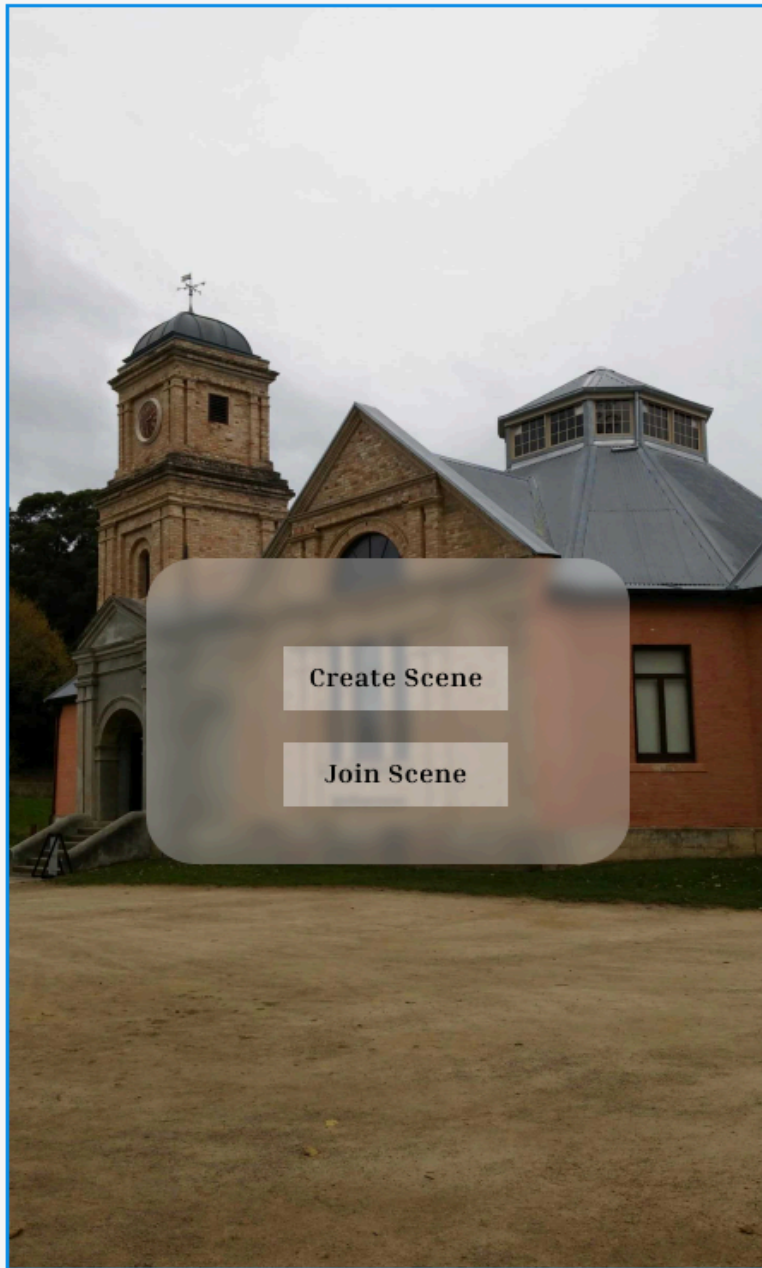
Crime scene View

The living room background simulates the headset being in an XR environment, and the brightly colored wardrobe simulates the insertion of a virtual 3D model.



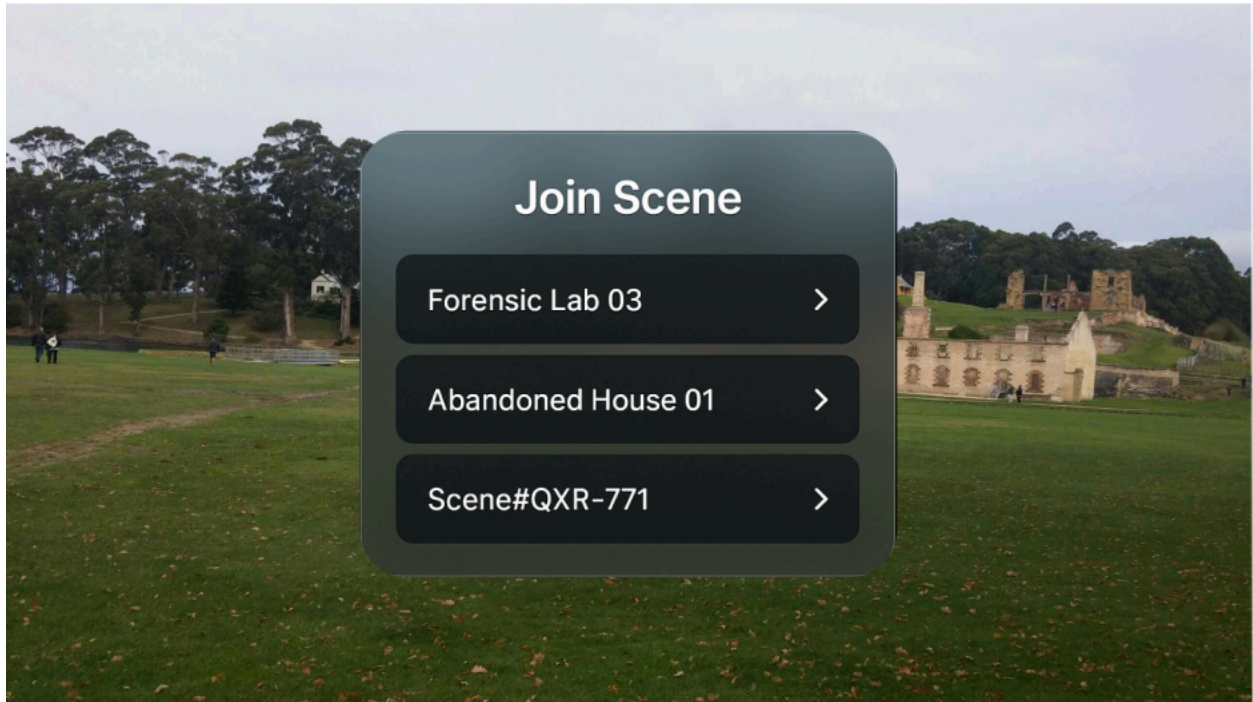
Menu Page_Main Page

The background house simulates the scene where XR passthrough is effective. The frosted glass effect panel simulates the effect of the virtual menu floating in the environment. The menu mainly includes creating a host scene or joining an already created scene.



Menu Page_Join Scene Page

The background environment simulates the scene where XR passthrough is effective. The frosted glass effect panel simulates the effect of the virtual menu floating in the environment. The menu mainly includes clicking join Scene in the first-level menu to enter the second-level menu and select to enter the created scene.



Welcome Page

The logo welcome page that appears first when entering the software



Feature/Function Test Plan

Feature ID	Feature	Test Case	Expected Result	Test Status
F001	Welcome Screen	Welcome Screen Display	Logo displays clearly at launch. Screen transitions automatically after a set amount of time.	Pending
F002	Session Management	Create Session	Option to create a new session available. Room ID is generated and clearly displayed.	Pending
F002	Session Management	Join Session	Users can input session ID. Successful session connection when valid ID entered.	Pending
F003	Scene File Loading	File Selection	File selection prompt after session creation.	Pending
F004	Scene Alignment	QR Code Scanning	QR scanning prompt clearly shown. Successful QR scan correctly anchors the scene.	Pending
F005	Scene Interaction	Scene Manipulation	Users can move, rotate, and scale scenes intuitively. All interactions smoothly synchronised in real time.	Pending

Candidate Tests

Test Case:File Selection

Test Case	File Selection	
Feature Id	F001	
Feature	File Selection	
Test Purpose	To ensure that the user is able to load and render FBXor OBJ files into the scene.	
Screen Ref	Welcome Screen and File Selection	
Expected Results	Users are able to load the fbx/obj file from local storage and the selected file is rendered correctly in the scene.	
Test Status	Passed	
Test Case Date		
Field Name	Set 1	Set 3
File Name	Scene1.fbx	Scene2.obj
Expected Result	Pass	Pass
Status	Pass	Pass
Date Tested	17/04/2025	17/05/2025
Action to be Taken	-	-

Test Case: Session Management

Test Case	Session Management	
Feature	F002	
Feature	Manage Session	
Test Purpose	Validate creation of new sessions and joining of active sessions.	
Screen Ref	-	
Expected Results	Users are able to create a session or join an active session. All the users synchronized to the same scene location	
Test Status	Pending	
Test Case Date		
Field Name	Set 1	Set 2
Session type	Create session	Join session
Expected Result	Pass	Pass
Status		
Date Tested	Yet to be tested	Yet to be tested
Action to be Taken	-	QR code validation required

Test Case: QR Code Scanning

Test Case	QR Code Scanning	
Feature	F003	
Feature	QR Code Scanning	
Test Purpose	Ensuring the origin is set to (0,0,0) after the QR code is scanned	
Screen Ref	QR Code Scanning Interface	
Expected Results	Spawn scene at the correct location	
Test Status	Pending	
Test Case Date		
Field Name	Set 1	
QR Code Validation	Valid QR code	
Expected Result	Pass	
Status		
Date Tested	Yet to be tested	
Action to be Taken	Error handling for invalid QR codes	

Test Case: Scene Manipulation

Test Case	Scene Manipulation	
Feature	F004	
Feature	Scene Manipulation	
Test Purpose	Validate that the users can move and rescale the LiDAR scene effectively.	
Screen Ref	Scene Manipulation Interface	
Expected Results	Users are able to reposition the scene and rescale the scene, with changes and synchronize across all devices	
Test Status	Pending	
Test Case Date		
Field Name	Set 1	Set 2
Scene Size	Large Scene	Small Scene
Expected Result	Pass	Pass
Status		
Date Tested	Yet to be tested	Yet to be tested
Action to be Taken	-	-

Test Case: Passthrough and Depth Sensing

Test Case	Passthrough and Depth Sensing	
Feature	F005	
Feature	Passthrough and Depth Sensing	
Test Purpose	Ensuring safety navigation and image stability using passthrough and depth sensing technologies.	
Screen Ref	Passthrough and Depth Sensing Interface	
Expected Results	Users are able to navigate in the physical environment and visuals are stable during exploration	
Test Status	Pending	
Test Case Date		
Field Name	Set 1	Set 2
Scene Size	Large Open Area	Area with Obstacles
Expected Result	Pass	Pass
Status		
Date Tested	Yet to be tested	Yet to be tested
Action to be Taken	-	Improve Obstacle Detection

Conclusion

This System Analysis and Design Report delivers detailed descriptions and operational features of the LiDAR Crime Scenes for VR project that operates on the Meta Quest 3 platform. A complete documentation process ensured system requirements through use case modelling, acceptance criteria development, detailed testing to deliver functional and usable and performant systems.

The project executes essential features which handle session management and enable precise scene loading together with QR code-based scene alignment as well as interactive manipulation capabilities and safety navigation through passthrough and depth sensing technology. The system includes selected features which meet user requirements to create secure collaborative environments that improve the investigative process for forensic specialists.

The system identifies potential issues and provides detailed prevention methods to manage successful deployment across the user experience framework. The modeling methods combined with comprehensive planning situated in this design report create a solid base for successful implementation which delivers immediate results along with sustainable growth potential. This proposed solution brings substantial value through technological innovation which allows complete support for modern forensic investigation objectives.

References

Western Sydney University (Ed.). (2025). *Western Sydney University 2025, Project Brief – PA2520: LiDAR Crime Scenes for VR*.

Western Sydney University 2025 (Ed.). (2025). *COMP3018 Professional Experience – Assessment Guide, Western Sydney University*.

Dennis, A., Wixom, B. H., & Roth, R. M. (2009). *Systems Analysis and Design*. Wiley.
https://www.google.com.au/books/edition/Systems_Analysis_and_Design/gFkYqK84rVgC?hl=zh-CN&gbpv=1

MATHA, M. P. (2008). *Object-Oriented Analysis and Design Using UML: An Introduction to Unified Process and Design Patterns*. PHI Learning.

https://books.google.com.au/books?id=z5XX4gYuMs0C&printsec=frontcover&hl=zh-CN&source=gbps_ge_summary_r&cad=0#v=onepage&q&f=false